TEAM CODE CONQUERORS
Sardar Patel Institute of Technology, Mumbai

AI-Powered Financial Analysis tool for Indian Markets

Aarushi Ghosh
Aryan Shewale
Vivek Yadav
Vidhi Gond

CHALLENGES & PROBLEM STATEMENT

CHALLENGES

01
UNSTRUCTURED & FRAGMENTED DATA

Critical insights are scattered across PDFs, transcripts, slides, and filings with no unified structure.

02
MANUAL EXTRACTION BOTTLENECKS

Analysts spend hours identifying ratios, projections, and risk disclosures.

03
TIME-TO-INSIGHT LAG

Market-moving signals are often identified too late due to manual processing delays.

04
MODEL-BUILDING INEFFICIENCY

Financial modeling requires repetitive data cleaning, validation, and cross-referencing.

01
LIMITED SCALABILITY

Deep fundamental analysis cannot be efficiently replicated across hundreds of small- and mid-cap companies.

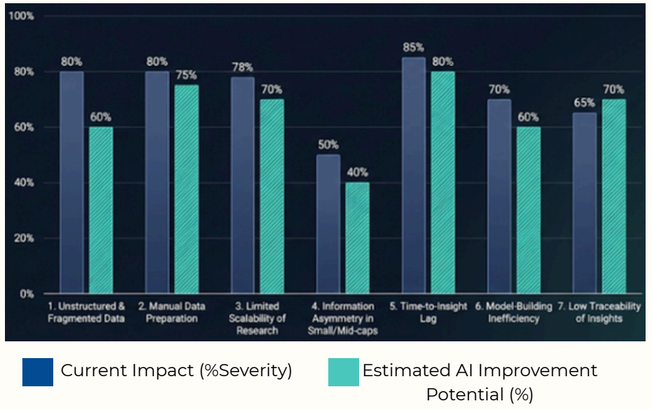
PROBLEM STATEMENT

India's equity markets are witnessing unprecedented participation from both retail and institutional investors, especially in the rapidly growing small- and mid-cap segments. However, despite the surge in capital and activity, critical financial intelligence remains buried within vast volumes of unstructured corporate disclosures - including investor presentations, earnings call transcripts, regulatory filings, annual reports, and management commentary.

While rich in insight, this data is fragmented, difficult to process, and hard to access at scale. Analysts and investors must manually extract key metrics, interpret guidance, assess risks, and build forward-looking financial models - a process that is time-consuming and inefficient.

In today's AI-driven world, this creates a widening gap between available data and actionable intelligence.

CHALLENGES IN TRADITIONAL EQUITY RESEARCH



SOLUTION OVERVIEW

AlphaSage is an **AI-powered** equity research platform redefining **financial intelligence** in the Indian markets through a **scalable**, **multi-agent** architecture that delivers **real-time**, investment-grade insights. Built on AutoGen (v0.7), it seamlessly integrates **MongoDB**, **ChromaDB**, **yfinance**, and **Gemini 1.5 Pro** / **Gemini 1.5 Flash** to enable advanced contextual reasoning and structured financial analysis.

ITS THREE-TIER AGENT ARCHITECTURE INCLUDES:

Micro Agents for **precision** tasks such as **P/E extraction**, **ratio computation**, and **sentiment** tagging

Macro Agents for higher-order **intelligence** including revenue **forecasting**, **risk mapping**, and strategic

Output Agents (upcoming) to **automatically** generate dashboards, research reports, and financial models

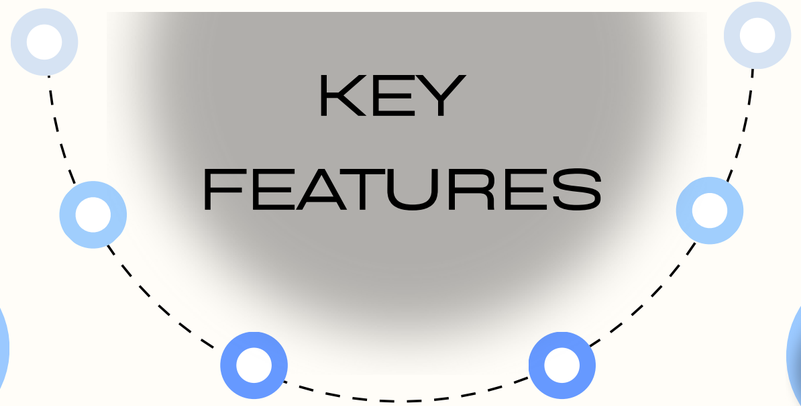
Built for **speed** and **scale**, AlphaSage unifies **advanced caching**, **asynchronous** processing, and **optimized** AI orchestration with complete **source traceability**. It transforms **fragmented** disclosures into **precise**, **decision-ready** insights - **delivering faster**, **sharper**, and more **transparent** equity intelligence.

KEY FEATURES

Multi-Agent Intelligence

Three-layered architecture powered by AutoGen (v0.7+):

- **Micro Agents**
- **Macro Agents**
- **Output Agent**



Future-Ready & Market-Aligned

- **Automated** dashboards, reports, and **Excel models**
- **Scalable** across companies and sectors
- Built for **transparent**, AI-driven **alpha discovery**.

Real-Time, Multi-Source Integration

- **Live** market data via yfinance **APIs**
- **OCR**-processed disclosures indexed in **ChromaDB**
- Semantic retrieval using **advanced embeddings**
- Supports event-driven **financial intelligence**.

Scalable, Cost-Efficient Processing

- **Token-optimized** pipeline using Gemini **1.5** Pro / Flash
- Structured chunking for **efficient** inference
- Async batch execution for **large**-scale **document handling**

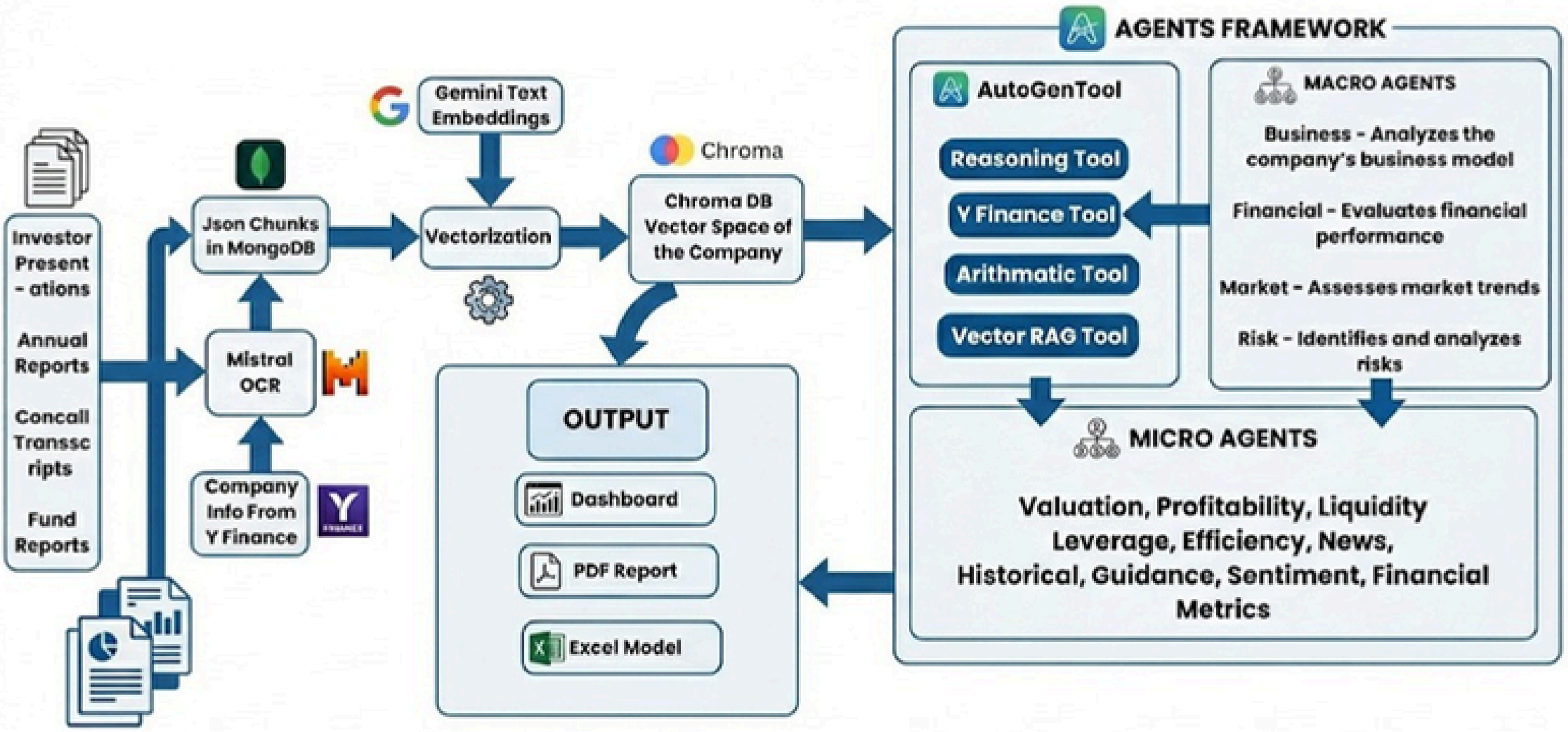
Traceability & Reliability

- Fully source-**linked** insights
- **Comprehensive** logging for auditability
- **Validation** layers with **automated** retry logic.

Modular & Developer-Friendly

- **Centralized** tool abstraction layer
- **Extensible**, clean architecture
- Plug-and-play agent **orchestration**.
- **API-ready** design enabling seamless integration

SYSTEM ARCHITECHTURE



TEAM ACCESS DENIED

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THANK YOU

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